

Exploratory Data Analysis Report

Zomato Business Intelligence & Delivery Time Prediction Platform

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1. Summary of findings

After cleaning, the dataset holds **20,475 orders** from **9,827 customers** across 1,200 restaurants. **13,544** orders were completed (Delivered or Delivered Late) generating **INR 53.1 lakh** (average order INR 392). The most important facts are operational: **22.3% of orders are cancelled and a further 11.5% are marked 'Food Not Delivered'** (so 33.9% fail), and **15.9% of completed deliveries are late**. Average delivery time is 38.3 minutes (median 36, p90 60).

The second important fact is a **null result**. We ran 23 permutation tests asking whether cancellation, lateness, delivery time or order value differ by city, cuisine, weather, traffic, payment method, vehicle, hour or membership. Only 2 reached $p < 0.05$; with 23 tests about 1.2 are expected by chance alone. Differences visible in the charts (e.g. 'Mexican has the highest cancellation rate') are therefore **not distinguishable from random variation** and should not drive decisions.

What is structured: ordering time (lunch 12-14h and dinner 19-21h peaks, identical on weekdays and weekends), the late-delivery group (mean 63.7 min vs 33.5 min for on-time), and the monthly revenue level (flat at about INR 2.2 lakh/month).

2. Data cleaning (PRD 19.1)

All 52 logged transformations are in *data/cleaned/cleaning_log.csv*; the cleaned schema passes 31/31 integrity checks (unique primary keys, all foreign keys resolve, no negative costs, ratings 1-5, promotion end $\geq$ start).

Three decisions were tested on the data rather than assumed:

- **Negative values are sign flips.** For all 181 orders with negative FoodCost, $|\text{FoodCost}| + \text{DeliveryFee} - \text{Discount} + \text{GST}$ reproduces FinalAmount exactly, so we use `abs()` rather than drop the rows. The same rule is applied to delivery time, price, age, humidity and speed.
- **Mixed day/month order in slash dates.** Within one column both DD/MM/YYYY and MM/DD/YYYY occur (about 15% of dates are ambiguous). For orders we pick the reading closest to the same order's payment date; weather/traffic default to DD/MM.
- **Weather/traffic dates repeat.** Each city has ~730 rows but only ~63% of calendar days, so (City, Date) is not unique and only ~63% of orders link to weather/traffic. Missing links are imputed with the city median and flagged (HasWeather / HasTraffic).

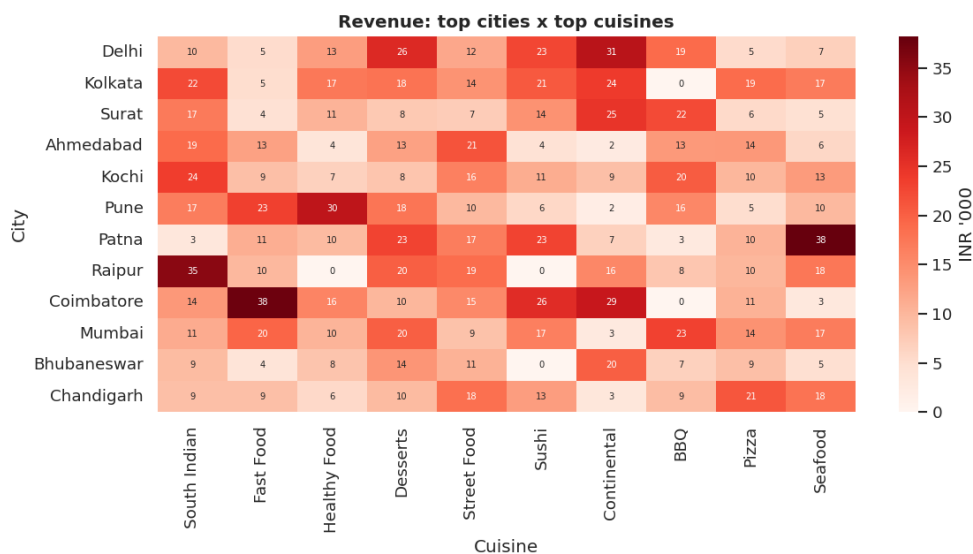
table	issue	column	rows
orders	ambiguous slash date	OrderDate	3189
weather	ambiguous slash date	Date	2856
traffic	ambiguous slash date	Date	2797
orders	ambiguous date flipped	OrderDate	1727
customers	ambiguous slash date	RegistrationDate	1539
customers	city spelling/case/whitespace	City	1236
customers	invalid/missing phone	Phone	592
order_items	missing	TotalPrice	435
order_items	negative value (sign flip)	Quantity	417
payments	multiple payments per order	OrderID	399

Largest cleaning actions by rows affected (date parsing excluded; every row of each date column is parsed).

3. Business questions

Q1. Which cities, cuisines and restaurants earn the most, and how has revenue changed?

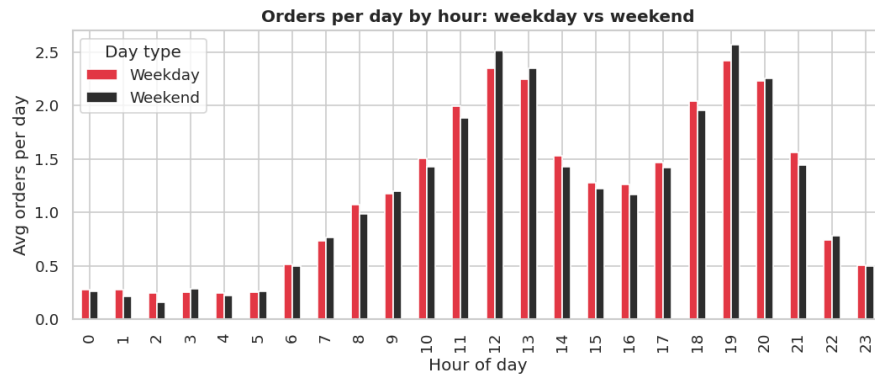
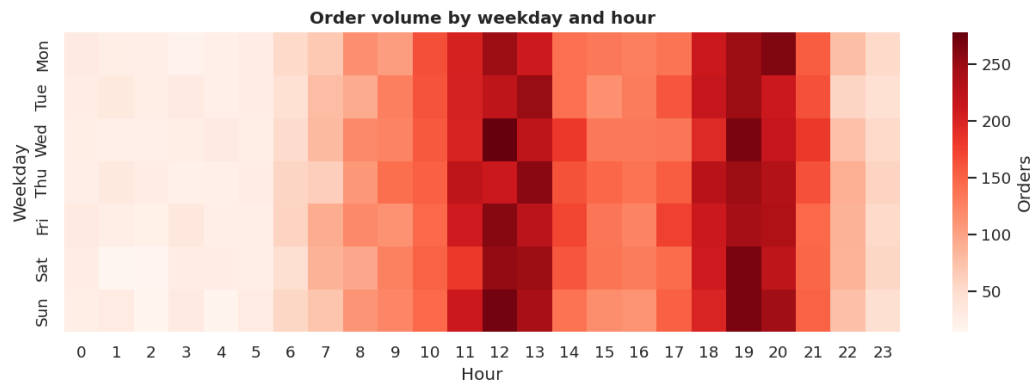
Top cities by revenue: Delhi (2.47 lakh), Kolkata (2.40 lakh), Surat (2.39 lakh), Ahmedabad (2.37 lakh), Kochi (2.35 lakh). Top cuisines: South Indian (3.68 lakh), Fast Food (3.36 lakh), Healthy Food (3.13 lakh), Desserts (3.00 lakh), Street Food (2.95 lakh). The spread between the #1 and #5 city is under 5%, inside what random assignment of orders would produce. Monthly revenue is flat (INR 1.9-2.5 lakh; first-12-month total 26.6 lakh vs last-12-month total 26.5 lakh) - there is no growth trend.



City x cuisine revenue (INR '000). No block stands out beyond sampling noise.

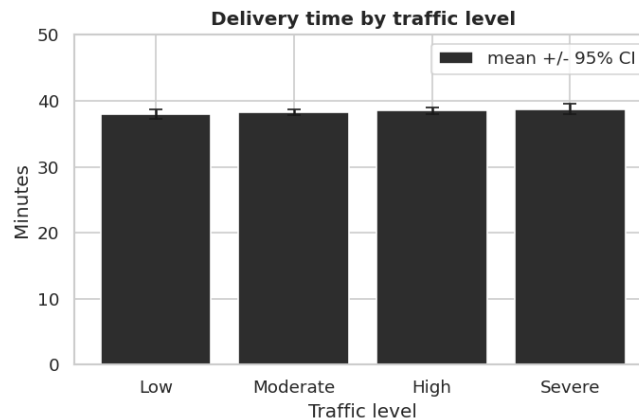
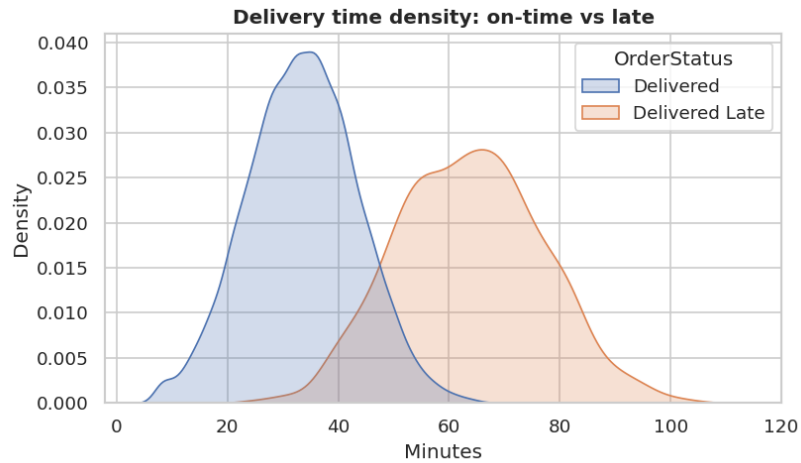
Q2. When do people order?

Orders peak at 19:00 with a clear lunch (12-13h) and dinner (19-20h) structure. Weekends hold 28.3% of orders (28.6% expected if days were equal), i.e. no weekend uplift. This is the one pattern in the data with strong structure, and it supports the PeakHour feature.



Q3. How are delivery times distributed and what moves them?

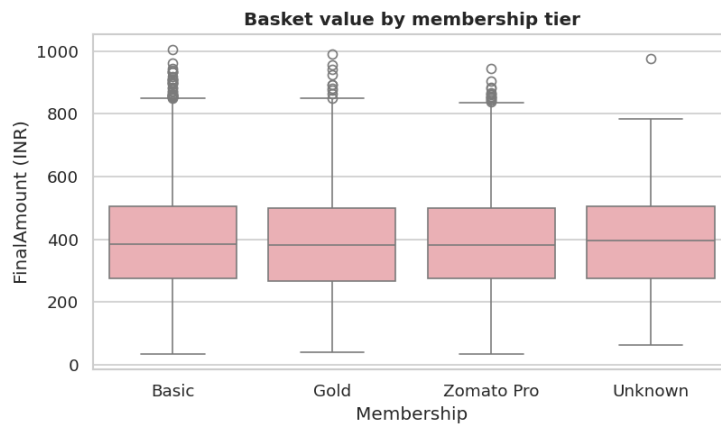
Median 36 min, p90 60, p95 70. Average time by traffic level (Low to Severe) is 38.0, 38.3, 38.5, 38.7 minutes - a total spread of 0.7 minutes (permutation $p = 0.36$). By rain: 38.0, 38.4, 38.2, 38.7 ($p = 0.47$). City ($p = 0.18$), vehicle ($p = 0.43$) and weather condition ($p = 0.16$) are also not significant. The distribution is bimodal: 'Delivered Late' orders average 63.7 min versus 33.5 min.



Mean delivery time by traffic level with 95% CI: bars overlap completely.

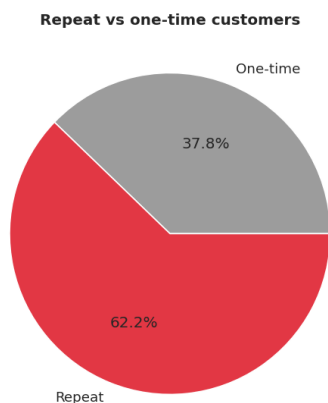
Q4. What is the average basket and does it vary by membership?

Average order value is INR 392. By membership: Basic 394, Gold 390, Unknown 395, Zomato Pro 389 - no tier effect ($p = 0.84$). Orders with a coupon average INR 324 versus 429 without; this gap is largely arithmetic because FinalAmount is net of the discount, so it says nothing about whether coupons grow baskets.



Q5. Repeat versus one-time customers

37.8% of customers who ordered did so exactly once; in addition 2,173 of 12,000 registered customers (18.1%) never ordered. Orders per customer range 1-8 (mean 2.1).

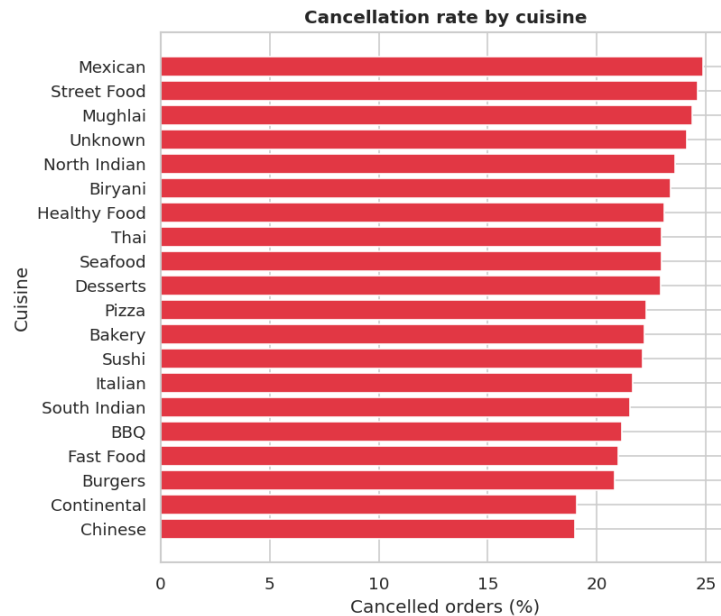


Q6. Which delivery partners are fastest and best rated?

Among 287 partners with at least 10 completed deliveries, the spread of partner average times is only marginally larger than chance (permutation $p = 0.09$), so 'fastest partner' rankings are mostly noise. Partners need many more deliveries each before a leaderboard is trustworthy (see Power BI Delivery page, which shows the minimum-volume filter).

Q7. What drives cancellations?

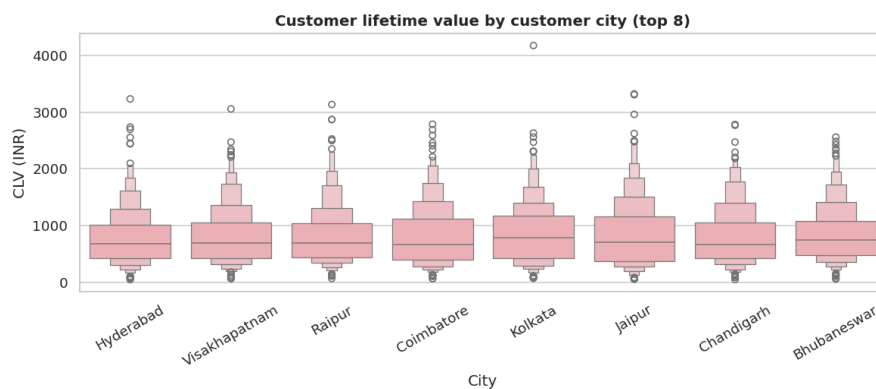
22.3% of orders are cancelled. Cancellation by cuisine ranges from 19.0% to 24.9% (chi-square permutation $p = 0.058$, borderline); by payment method $p = 0.036$ (1 of 23 tests - consistent with chance); by city $p = 0.95$; weather $p = 0.83$; hour $p = 0.84$; coupon use $p = 0.29$. Restaurant rating is unrelated to a restaurant's cancel rate ($r = 0.014$).



Cuisine differences are within sampling noise (borderline $p = 0.06$; one of many comparisons).

Q8. How is customer lifetime value distributed?

CLV is the sum of completed-order value per customer; it is right-skewed with most customers below INR 800 because the median customer places 2 orders. City-level differences are small relative to the within-city spread.



4. Data-integrity observations worth a data-owner review

- **Payment status is independent of order status.** About 80% of cancelled and 'Food Not Delivered' orders show payment 'Success' and only ~8% 'Refunded' - the same split as delivered orders. Either the two fields are not synchronised or refunds are not recorded.
- **Feedback does not reflect lateness.** Mean delivery rating is 3.51 for late and 3.47 for on-time deliveries, and 'Delivery took way too long' style reviews are spread evenly across statuses.
- **Customer and restaurant cities agree on only ~4% of orders,** and restaurant coordinates do not match their cities, so no real delivery distance exists. The 'DistanceProxyKm' feature is a proxy only.
- **Cancelled orders have delivery times** (mean about 35 min, same as delivered), which a real system would not record; delivery-time modelling therefore uses completed orders only.

5. Charts

All 38 charts carry titles, axis labels and legends where relevant; see *images/* (eda_mpl_* and eda_sns_*). The correlation heatmap below shows that no numeric feature correlates with delivery time ($|r| < 0.03$).

